The Brain and Heart – Disturbed: A Predictable Pair or Dichotomized Duo?

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Abstract

From the early days of functional neuroimaging, researchers have leveraged peripheral physiological sensing and concurrent use of the technology to explore brain-heart connections. From demonstrating simple correlations between parameters such as heart rate and fear-implicated brain regions such as the amygdala to very recent prediction of heart rate and skin conductance using fMRI data, a vast amount of literature has established that at least in certain contexts, limbic brain activity and certain peripheral physiological features are significantly intertwined. Yet, translating this scientific result to an application-oriented view of using peripheral biosignals to estimate brain activity has not been accomplished. This project focused on this regression problem to lay the framework for future research in this area. Due to data limitation constraints, however, we sought to find clever ways to summarize our plentiful physiological data to better map to brain activity. One of these ways was inspired by LFADS, in that we hypothesized that latent autonomic nervous system state dynamics drive the observable changes in peripheral physiological features observed, and hence, by constructing such a nonlinear latent state model, we may be able to use lower dimensional factors to map to brain activity. To this end, we experimented with a variety of subject-independent approaches to this regression problem using LSTMs, support vector regression, ridge regression, etc. Concurrently, we implemented a many-to-one LSTM as a fully supervised approach for comparison. Upon observing unsatisfactory results from all subject-independent approaches, we spent the remaining couple weeks of the semester stepping back and analyzing the data using simpler correlation analysis, as well as by experimenting with subject-specific modeling. These results were more promising, with statistically significant correlations being discovered and positive R squared values being produced for subject-specific regressions. These results give us hope for fruitful research in this area, even if the exact ideas laid out herein prove intractable.

Background and Introduction

Unhealthy levels of distress can affect the body through a myriad of autonomic nervous system (ANS) pathways, including cardiovascular changes that increase cardiac output (e.g., increased heart rate), respiration changes to increase minute ventilation (e.g., increased respiration rate), and variations in limbic and brainstem activity that mentally and emotionally focus the individual on the stressor at hand [1], [2]. Each of these responses plays a pivotal role in preparing the body for “fight or flight,” ensuring enough energy can be wielded at a moment’s notice to ensure survival. Yet, when this system goes awry, the body can inadvertently remain on high alert for little to no good reason. In humans, well-known examples of this neuropsychological dysregulation include trauma and anxiety disorders, such as PTSD or panic disorder, where the equilibrating balance between sympathetic and parasympathetic activity is left compromised, and sympathetic hyperarousal can overtake one’s life [3], [4].

The Inan Research Lab, in collaboration with PTSD expert J. Douglas Bremner, MD at Emory, has collected a dataset investigating the effects of transcutaneous cervical vagus nerve stimulation on peripheral biosignals and cerebral blood flow in the context of traumatic and mental stress in subjects affected by prior trauma [5], [6]. Data from both PTSD and non-PTSD subjects were collected; however, note that for this course project, we focused on non-PTSD subjects because their neuroimaging data are

relatively cleaner. The protocol involved an approximately two-hour-long data collection where four peripheral cardiovascular and respiratory signals were measured throughout: electrocardiogram (ECG), photoplethysmogram (PPG), seismocardiogram (SCG), and respiratory effort (RSP).

ECG measures the electrical activity of the heart and is the well-known signal used to extract cardiac polarization and depolarization information. SCG measures the mechanical activity of the heart and is used to assess the subsecond-level delay between left ventricular depolarization and the ejection of blood to the rest of the body. PPG is also known as the blood volume pulse signal and is used to extract a variety of information related to blood flow through peripheral arteries (e.g., volume of blood flow, affected by vasodilation/vasoconstriction and pulse pressure). RSP is measured using a simple belt that stretches and compresses in response to breathing and is used to measure respiratory timing.

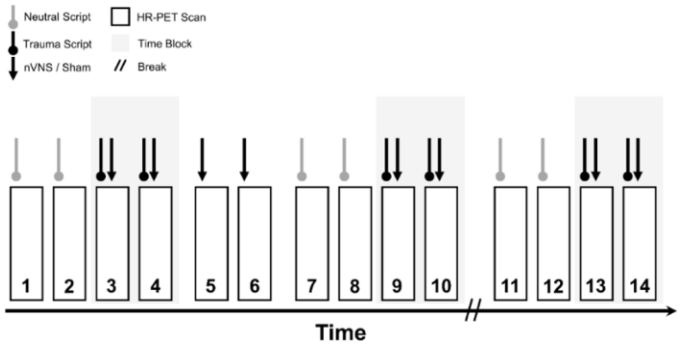

Figure 1. HR-PET Scan Protocol Diagram

In conjunction with this non-invasive biosensing, ten high-resolution positron emission tomography (HR-PET) brain images were collected over the span of this period. These scans were taken in coincidence with stimuli such as traumatic scripts, neutral scripts, and transcutaneous cervical vagus nerve stimulation. Please see Figure 1 for a simplified timeline of the protocol. The break between scans 10 and 11 was a lunch break; hence, to ensure consistency in baseline for this project, we only considered data from before lunch. Note that each HR-PET scan takes a fixed 120 s to produce one functional cerebral blood flow image, i.e., one value is obtained for each voxel over the span of the entire 120 s.
Collaborators from Emory had already processed the HR-PET data and extracted blood flow measures (continuous values) for specific limbic regions in the brain (left and right amygdala, anterior cingulate, mid cingulate, hippocampus, ventromedial prefrontal cortex, insula, and thalamus, for a total of 14 brain areas). To keep things tenable, we only considered the right amygdala for much of this project's analysis, so these data were readily available to the group from project onset.

Unlike prior work that has restricted analysis to single digital biomarkers at a time (including his paper referenced above), Asim's current efforts focus on multimodal fusion in an attempt to map these observable markers back to an underlying autonomic nervous system "stress" state. To this end, 11 time series features and 12 variability features (variability features were extracted in moving window fashion and were treated as additional time series) had already been extracted (resampled to 1 Hz) from the physiological signals prior to the start of the project and were hence readily available for the group.
Table 1 lists these features. Each of these quantities is known to be modulated by sympathetic activation or parasympathetic deactivation (i.e., the prototypical stress response), and these time series features, in conjunction with limbic brain activity from the 8-10 brain images collected per subject, served as the data we explored for the project.

Because we hypothesize that the observable, yet seemingly more complex, stress-related changes in physiology are projections of latent lower-dimensional ANS state dynamics, the first component of this

course project involved taking a step toward computationally instantiating this hypothesis. Specifically, we attempted to model the time series data we observed with a nonlinear latent state dynamical modeling approach inspired by LFADS. The second component of the project that maybe does more to solidify the connection of this project to the "Neuroscience" part of this "Quantitative Neuroscience" course has to do with an additional hypothesis: namely, that these latent low-dimensional dynamics, which we propose represent the ANS modulating the periphery via its numerous sympathetic and parasympathetic pathways, are inherently related to limbic and brainstem activity. Given the well-established correlations between a number of the features and brain areas listed above [7]–[10], we believe that this hypothesis is not too much of a stretch, though demonstrating its validity, or invalidity for that matter, will take a sizable amount of work. This is due to the inherent difficulty in working with these peripheral measures and the multiple computational steps that need to be implemented and iterated upon using this data-driven approach. Not to mention, the data are not vast.

Signal	*Features*
Electrocardiogram (ECG)	Heart rate and the following heart rate variability features: root mean square of successive differences (RMSSD), proportion of normal-to-normal R-R intervals that differ by more than 50 ms (pNN50), low frequency:high frequency ratio (LF/HF), poincare-plot-based standard deviation 1: standard deviation 2 (SD1/SD2), respiratory sinus arrhythmia (RSA)
Seismocardiogram (SCG)	Pre-ejection period, reciprocal of pulse transit time
Photoplethysmogram (PPG)	PPG amplitude and pulse arrival time
Respiratory Effort (RSP)	Respiration rate, inspiration time, expiration time, inspiratory:expiratory ratio, and the RMSSD, coefficient of variation, and autocorrelation at one breath lag for each of the first three RSP features mentioned for respiration pattern variability

Table 1. List of peripheral biosignals and features

However, with all this pessimism laid out, we optimistically note that the amount of concurrently collected data are not zero. We also highlight the fact here that this "heart to brain" regression problem is one that has not yet been published upon to the best of our knowledge. Although the correlations aforementioned are promising, no one has shown the ability to estimate on held-out data in a sort of "wearable brain activity estimator" application. Hence, the potential for impact is large. The most similar work in this sense would be the work of Eisenbarth et al. [11], where the predictive connection was made in the opposite direction, i.e., fMRI data were used to predict skin conductance and heart rate changes. We believe this lack of literature exploring the opposite direction of this regression problem may be due to the lack of physiological measures commonly extracted. Thus, because we have unique access to a diverse array of relevant physiological features and wanted to experiment with the ideas laid out thus far, we defined our project objectives as follows:

(1) Implement the aforementioned dynamical systems hypothesis computationally, with inspiration drawn from the newfound knowledge of work done by quantitative neuroscientists studying neuronal populations

(2) Experiment with this novel brain-heart regression problem, in conjunction with and separate from, the dynamical systems hypothesis made above, where physiological features are used to estimate activity in particular areas of the brain

Figure 2 illustrates the spirit of objective 2, where the work from objective 1 could potentially serve as a component of the functional mapping learned, f().

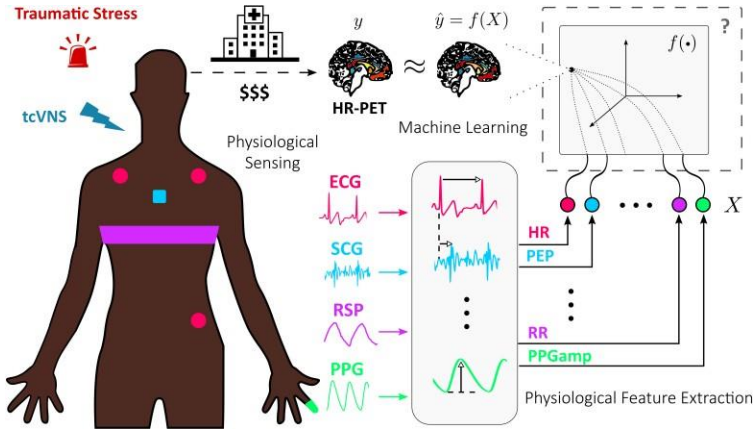

Figure 2. High-Level illustration of heart-to-brain regression problem

Methods

Feature Preprocessing

First, we performed interpolation (with sampling rate = 1Hz) on the features as they were not equally sampled in time. The interpolated features were smoothed using 5-point moving average. We performed normalization by subtracting the features from their 5-minute mean, and dividing them by this same mean. The rational for this normalization procedure is to extract just the changes of the features relative to the baseline. Through this procedure, the feature distributions were slightly more consistent across subjects.

Label Preprocessing

Labels associated with different brain regions were extracted from HR-PET data during each of the HR-PET scans (120s). The labels would indicate blood flow measures of different brain regions. For the purpose of this study, we only considered the right amygdala labels.

Before we trained our models, we normalized our labels via two different approaches. In the first approach, we set the second scan number in each subject as a baseline due to the consideration of stability. We subtracted features with the baseline mean and divided features with baseline standard deviation. As for the second approach, we used z-score as normalization.

Visualization

After preparing the time series and brain activity data for analysis, we visualized our data using factor analysis. This involved applying factor analysis to the physiological time series data corresponding to brain scan time windows and color coding the points based on the brain activity during that scan. To better understand the separability of the data between scans within subject and between subjects, we constructed a plot that included all the subjects' data, as well as a set of plots for each subject. We also applied principal component analysis to observe whether the results would differ significantly, but we found no apparent differences, so we only show factor analysis plots herein.

LFADS-Like Dynamical Autoencoder

To experiment with the dynamical systems hypothesis introduced earlier, we constructed an LFADS-like model to extract latent dynamical factors underlying our observed physiological data. In this way, we could leverage the plentiful physiological time series data to learn better representations of our data, without ever spending any of our precious brain activity labels.

The model was constructed using the same initial stages as LFADS. Namely, the encoder portion of the network consists of a bidirectional gated recurrent unit (GRU) that encodes the input sequence of physiological data into a single vector – this vector is the concatenation of the final GRU states from both GRUs (forward and backward). This larger vector is then fed into a linear feedforward network to reduce its size back down to the dimensionality of the hidden state. Now, because our analysis involved both an autoencoder (AE) and a variational autoencoder (VAE), the paths somewhat diverge here, but in both cases, the output of the encoder serves as some representation of the generator's initial conditions. For the VAE, instead of feeding in the concatenated output vector of the bidirectional GRUs directly into a single feedforward layer, this vector is fed into two separate feedforward layers. The first encodes the mean of the initial condition distribution, and the second encodes the natural logarithm of the variance of the initial condition distribution. For the VAE, this distribution is then sampled from to obtain initial conditions for the generator. In the AE case, there exists only one feedforward layer to directly output the initial conditions themselves, rather than having to do any sort of sampling.

The generator portion of the network consists first of a forward GRU that takes as initial state the initial condition from the encoder and as input at every time step the scalar zero, i.e., we reproduced autonomous LFADS. This GRU is then simulated forward to construct a sequence of vectors, representing the hidden state of the GRU at each time step. Each of these hidden state vectors are fed through a dimensionality bottleneck, where we differentiate the type of bottleneck that works on dimensionality from the bottleneck described thus far, where an entire sequence must be represented by a single vector / distribution and a trained generator. Here, the dimensionality bottleneck acts as any plain vanilla AE does. It takes each of the GRU states, reduces dimensionality down to the factor dimensionality using a linear feedforward network, and then decompresses the dimensionality directly into a reconstructed input. Here, we note the difference with LFADS in that we are not working with neural spiking data but rather with physiological feature time series, so we let this final linear feedforward network that represents the factor analysis transformation matrix output directly the reconstructed input.

The AE is trained using only the mean square error. A VAE, on the other hand, is trained using the sum of the mean square error, along with a hyperparameter times the KL divergence from a zero mean unit variance Gaussian distribution. For additional regularization beyond limiting the model capacity itself and sampling and KL divergence for the VAE, we used a 10% validation set to stop training based on the plateauing of the validation loss. Note that the reason we had to use an AE is that for subject-independent modeling, we found it quite difficult to produce satisfactory reconstructions using a VAE. With more tuning, we suspect that a VAE could work, but for the purposes of this project, we heeded the advice given after our presentation and tried an AE instead just so that reconstruction could at least be satisfactory in some sense (model can then focus on minimizing mean square error, regardless of how unwieldy the initial condition distribution becomes).

Subject-Independent Regression

In this section, we dealt with the problem using subject-independent approach. We stacked the data among all subjects directly and trained our models via leave-one-subject-out training, which means that we only tested on one subject each time.

Model 1: Simple LSTM model

Long Short-Term Memory (LSTM) is a specific recurrent neural network (RNN) architecture that was designed to deal with the vanishing gradient problem that may be encountered in conventional RNNs. With different gates, especially the forget gate, it can tell the cell state which information to forget or to keep, which is suitable for us to deal with a time series of data.

The input of a simple LSTM model is 3D tensor with shape (number of scans = 181, time points = 120, features = 23). The setting of the hidden layer is usually referred to the mature structure. Since there is no reference structure for the problem and the data in this project, the model was first designed based on experience in other fields and further determined by hyperparameter tuning. We only have a single hidden layer in our final setting with 16 units.

Model 2: Autoencoder + LSTM model

We hypothesized that we could extract more meaningful information via AE, so we also combined LSTM with autoencoder. To be more specific, instead of inputting the data we described before, we firstly passed our data into autoencoder, outputting three variables: factors, reconstructed data, and RNN states. Since most of the physiological responses triggered by stimuli were observed during the first 60 seconds, we selected a window length of 60 seconds for our AE. The input shape of factors, reconstructed data and RNN states are therefore (181, 60, 6), (181, 60, 23) and (181, 60, 20). Next, we fitted these three variables separately to our LSTM model and make the comparison with the performance from a simple LSTM model. The other parameters were the same as the simple LSTM model.

In this section, we first explained the datasets used in LSTM model as well as the data preprocessing, Next, we discuss the various LSTM NNs model for regression in our experimental design, including the combination of LSTM and autoencoders.

Implementation for subject-independent models

We implemented the models based on keras, a Python API designed for deep learning. To fully learn on the problem with sequence data, we extended traditional LSTMs to bidirectional LSTMs, which can provide additional context to the network. We use tanh activation function in cell recurrent connection whose second derivative sustain for a long span to address the vanishing gradient problem. We trained our model using Adam as an optimization algorithm with learning rate: 0.012 and mean square error as our loss function. In order to get a more intuitive evaluation, we also computed R square and included them into our results.

We also added L2 regularization with different parameters to avoid overfitting. The details will be described in discussion.

Mixed Model Correlation Analysis

Heeding the advice of classmates and instructors, we took a step back after unsuccessful attempts at subject-independent regression and took a look into simple correlations between the physiological features and activity within the limbic brain areas selected. The technical consideration made here was that repeated measures were made within subjects, but there were also multiple subjects' worth of data. To summarize the correlation results using something like a Fisher Transform would be a loss of statistical power, and to perform correlation analysis with the data as is, ignoring the repeated measures per subject, would be illegitimate given the dependence between various input datapoints due to the within-subjects variance inherent to the data.

Hence, we used a linear mixed-effects model, where the fixed effect for correlation analysis was the average of each of the physiological features considered, and the outputs were each of the brain activity datapoints. The random effect considered was the subject; to deal with these subject-specificities, we allowed for both the intercepts and the slopes to be different for each subject. 322 correlations were analyzed in this way, so to identify only pairs that truly had significant correlations, we applied a Bonferroni correction with significance level of 0.05, which resulted in necessary p-value of 1.55e-4 for statistical significance. Although we realize alternate methods are available such as Holm-Bonferroni correction, for the purposes of this project, we felt this was sufficient. Apart from this, we stored the p-values for all pairs and by ranking them smallest to largest, we were able to identify the strongest correlations with the right amygdala, the region that was focused on for the regression components of this project. This not only helped to inform the remainder of our subject-specific analysis, but also produced interesting results in their own right.

Subject-Specific Regression

Since subject variability can potentially hinder model performance, we also trained subject-specific models. To explore the methodology most useful for our problem, we applied ridge regression on three feature sets: all features ($X_{all} \in \mathbb{R}^{23}$), selected features ($X_{selected} \in \mathbb{R}^{5}$), VAE learned features ($X_{VAE} \in \mathbb{R}^{6}$). In particular, the $X_{selected}$ features (respiration rate, root mean square of successive differences in heart rate, proportion of normal to normal R peak to R peak intervals that differs by more than 50 ms, respiratory sinus arrhythmia, respiration pattern variability), were selected based on the preliminary feature/label correlation analysis.

Ridge regression was performed on these feature sets through a series of "leave-one-scan-out" (LOSO) cross validation (CV) experiments. For each subject, we spitted his/her data into K-folds by scan ID. Note that for VAE features, we only performed such feature extraction procedure on a single subject (subID 110) due to time constraints. We trained a regularized linear model on the training dataset. The $R^2$of the validation predictions was used to select the optimal regularization hyperparameter (HP) δ. We searched δ within the range of $10^{-15} to\ 10^{2}$ (100 values on logarithmic scale).

$$\underset{w}{\text{minimize}} \|Y - Xw\|_2^2 + \delta\|w\|_2^2$$

where Y is the label vector, X is the feature set, w is the weight, and δ is the regularization HP

$$R^2 = 1 - \frac{SS_{RES}}{SS_{TOT}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \overline{y})^2}$$

Where SS_{RES} Residual sum of squared errors and SS_{TOT} is the total sum of squared errors

Note that since we did not have sufficient scans (at most 10 for each subject), it is not reliable to perform a rigorous hyperparameter fine-tuning. Instead, we allowed some contamination of the labels through the HP optimization step. However, this experiment would provide valuable insights to the behavior of our data. The prediction results and R^2 for each feature set were reported.

Results and Discussion

Factor Analysis-Based Visualization

Figure 3 show subject-independent and subject-specific factor analysis plots, respectively. Recall that the color bars represent normalized brain activity. To streamline the results presentation for the remainder of this report, for subject-specific plots, we will show a select few and include the rest, if space allows, in the appendix. As an honest remark, we likely did not set aside enough time to work on

the report as the correlation and subject-specific regression results intrigued us enough to veer us off track from our original project timeline – we apologize in advance for the small font sizes in figures.

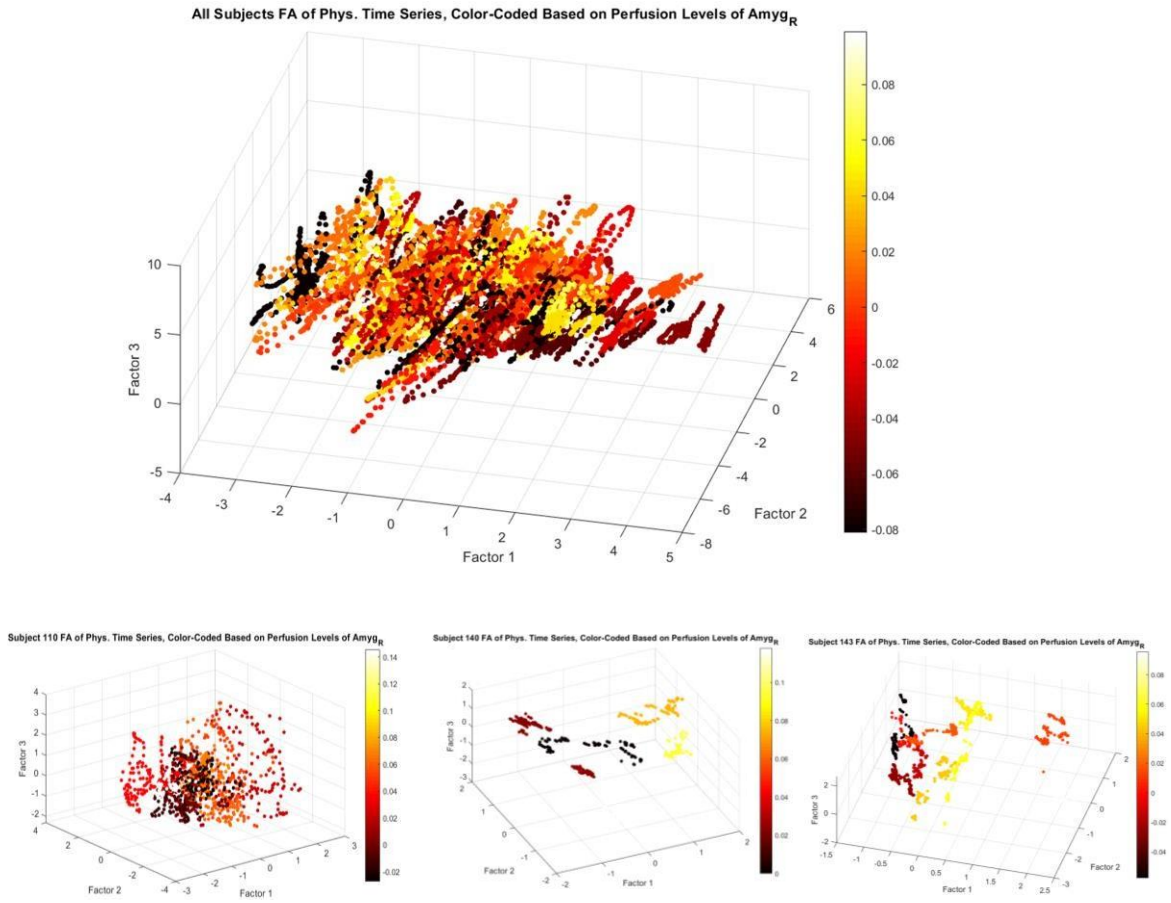

Figure 3. Factor analysis using 3 factors on all subjects (top), subject 110 (bottom left), subject 140 (bottom middle), and subject 143 (bottom right)

In the subject-independent factor analysis plot, we see that it is hard to decipher what we see. The sections of data corresponding to various brain activity levels within the right amygdala overlap significantly, and hence, not much can really be said. On the other hand, the subject-specific plots are quite intriguing. Subject 110's factor analysis plot seems disappointingly similar to the subject-independent plot, but the two subjects to the right have physiological data such that when factor analysis is applied for dimensionality reduction, distinct clusters appear according the particular scan. This can be observed by noting the clustering by color. That said, it does not seem like this low dimensional representation is fairly conducive to linear regression, as we see no apparent gradients. For the middle subplot, we do see some promising behavior where lesser brain activity values find themselves with smaller values of factor 1, while larger values find themselves being represented by the opposite; but the case on the right is more representative of our subject pool. Clustering is apparent, but it is difficult to conclude whether any region of the space represents higher or lower brain activity.

Subject-independent Regression

a. Subject 110 b. Best Scenario c. Worst Scenario

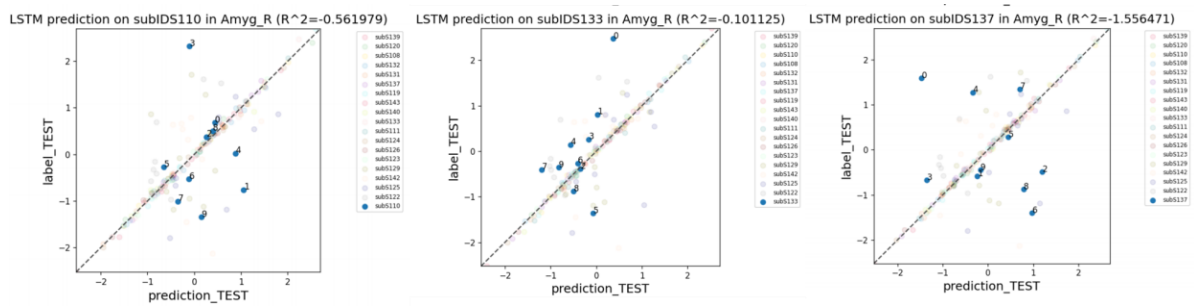

Figure 4. Prediction Performance via LSTM (L2 regularization)

a. Subject 110 b. Best Scenario c. Worst Scenario

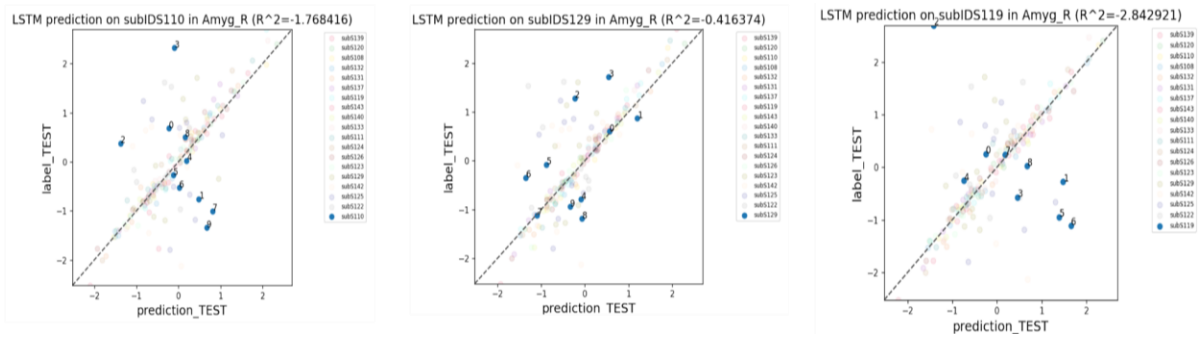

Figure 5. Prediction Performance via AE_LSTM with factors (L2 regularization)

In subject-independent models, we select subject 110 as a baseline to compare the results from different settings. In general, the performance via autoencoder + LSTM with factors as input data is worst. However, it took less time for training because the input shape is the smallest among these settings. Overall, the results using subject-independent are not good enough. It can at most achieve around the baseline model defined in R square. Even with the simple LSTM model can get the similar result. Nevertheless, we investigate that the core issue is not because of the architecture. Instead, the reason is from subject-independent setting. Therefore, we also deal with this problem with subject-specific setting in a following section.

Mixed Model Correlation Analysis

Figure 6 shows the statistically significant correlations between mean feature and brain activity, even after the conservative Bonferroni correction. Heart rate variability (HRV) features are well-known to possess important information related to ANS balance; thus, it makes sense to see two HRV features show up, the root mean square of successive differences in heart rate, which represents how variable heart rate is on a beat-by-beat basis, and the proportion of normal-to-normal ECG R-peak to R-peak intervals that differ by more than 50 ms. An interesting one that is worth noting even aside from this course project is the respiration pattern variability feature. Respiration pattern variability is only just recently becoming more mainstream as a feature indicative of “stress” or autonomic function. We find here that the autocorrelation of the respiration rate time series at one breath lag is significantly correlated with the left insula – an interesting brain region to be correlated with. The insula has been implicated in a number of activities, ANS regulation and consciousness to name a few. The fact that respiration has to do with both of these is certainly of interest!

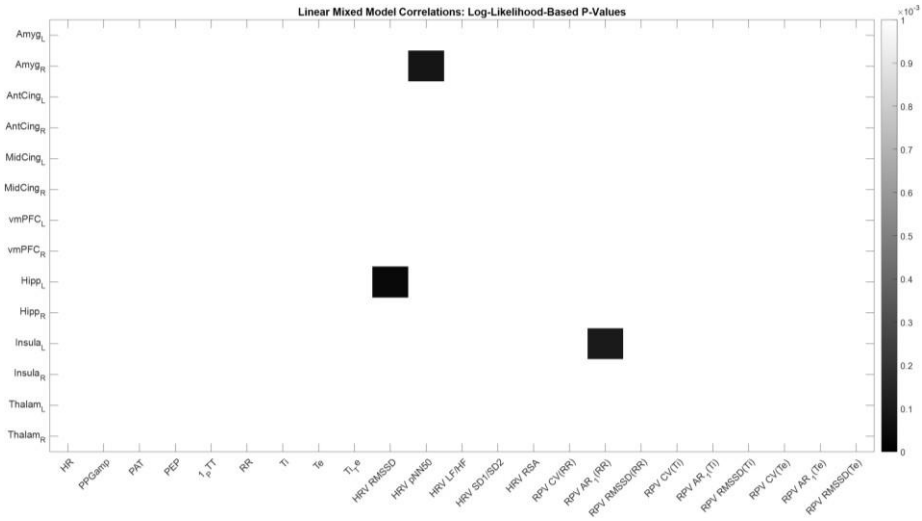

Figure 6. Statistically significant results from linear mixed model correlation analysis

Figure 7 shows all features that even have trend-level significant correlations (i.e., uncorrected $p < 0.05$). These features are listed in the subject-specific regression section, as we ended up using these features as one option for subject-specific regression. We see one more heart rate variability feature show up and one more respiration pattern variability feature show up.

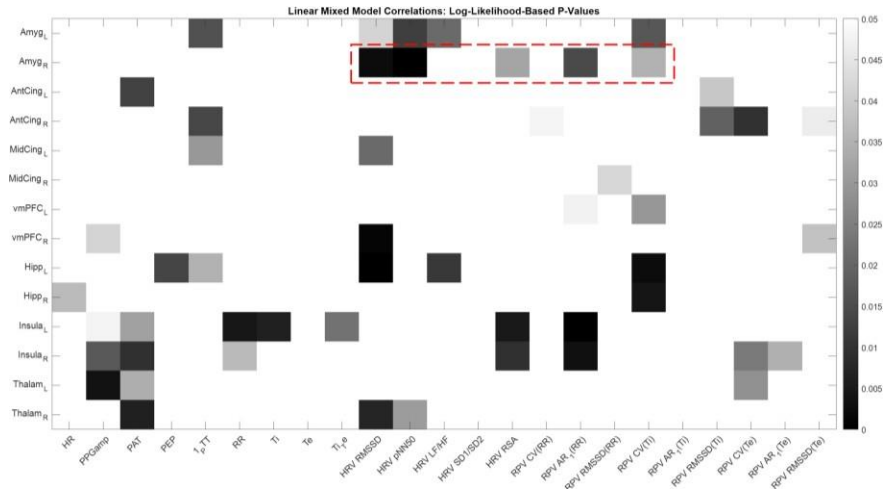

Figure 7. Features were selected based on trend-level correlations with right amygdala

Subject-specific Regression

a. Subject 110 b. Best Scenario c. Worst Scenario

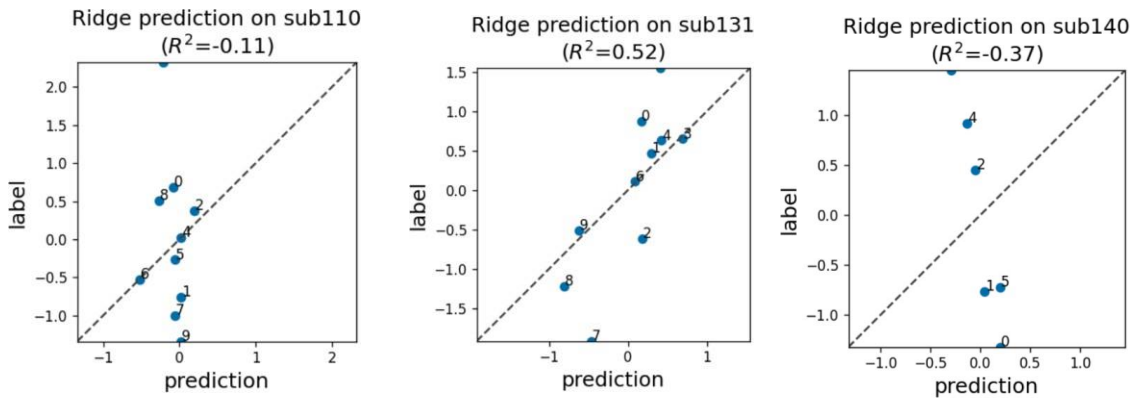

Figure 8. Ridge regression using all features

a. Subject 110 b. Best scenario c. Worst scenario

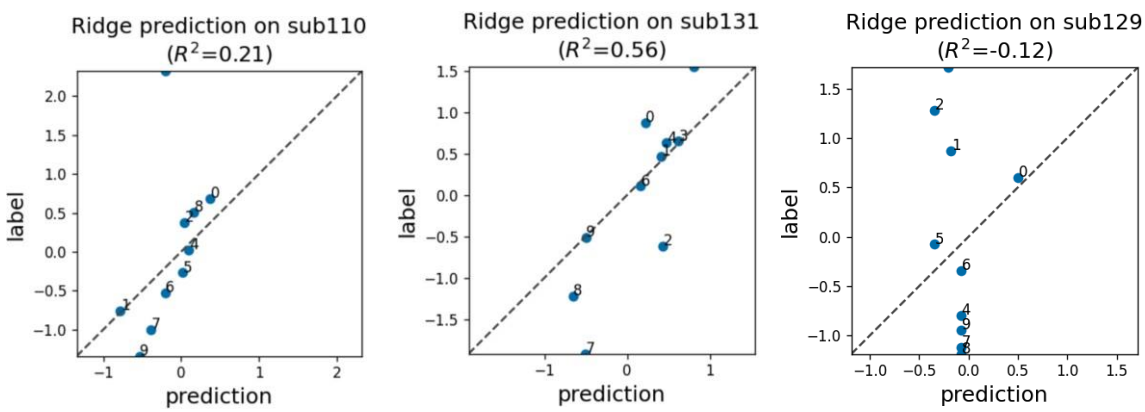

Figure 9. Ridge regression on selected features

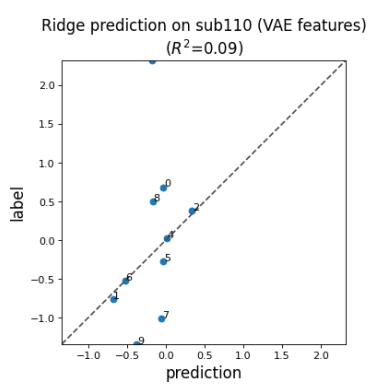

Figure 10. Ridge regression on features learned from VAE

The subject-specific models trained seemed to perform better than subject-independent models, indicating variability across subjects was hard to be removed with just LSTM or VAE combined with LSTM.

When consider just subject-specific models, we've examined three feature sets: X_{all}, $X_{selected}$, and X_{VAE}. When comparing just X_{all} and $X_{selected}$, we observed drastic improvements when model was trained on $X_{selected}$ across subjects, indicating the need to filter redundant/misleading features. Due to time constraints, we weren't able to train models on X_{VAE} for all subjects. Instead, we selected subject 110 (as the data showed promising results in the preliminary analysis). This model has also shown improvement in R^2 when compared to X_{all} trained model, though still inferior when compared to $X_{selected}$ trained model. Therefore, we believed our VAE was capable of filtering redundant information and extract compressed features, but it's likely information encoding stress level was also lost through this process.

Conclusion

Although unsatisfactory results from the initial subject-independent approaches plagued the initial steps of the project, we learned quite a bit through the iteration process, stepping back to assess correlations, and switching gears to subject-specific analysis. We found that thoughtful steps at every turn were key to producing even remotely promising results due to problem difficulty, but with enough experimentation, brainstorming, and implementation effort, certain analyses produced promising results, one of which would be of utmost interest to researchers in the brain-body field of research – namely, the correlation result between insula and respiration pattern variability. Additionally, the subject-specific factor analysis plots and regression results demonstrate that some promise remains with the ideas laid out in the Introduction section of the report. Thus, through this project, we not only obtained a better understanding of the problem and its difficulty through this preliminary analysis, but we also learned that even with the dataset as is, there is some hope to make both a scientific and an engineering impact if enough work is put into furthering the seed work accomplished here.

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Appendix

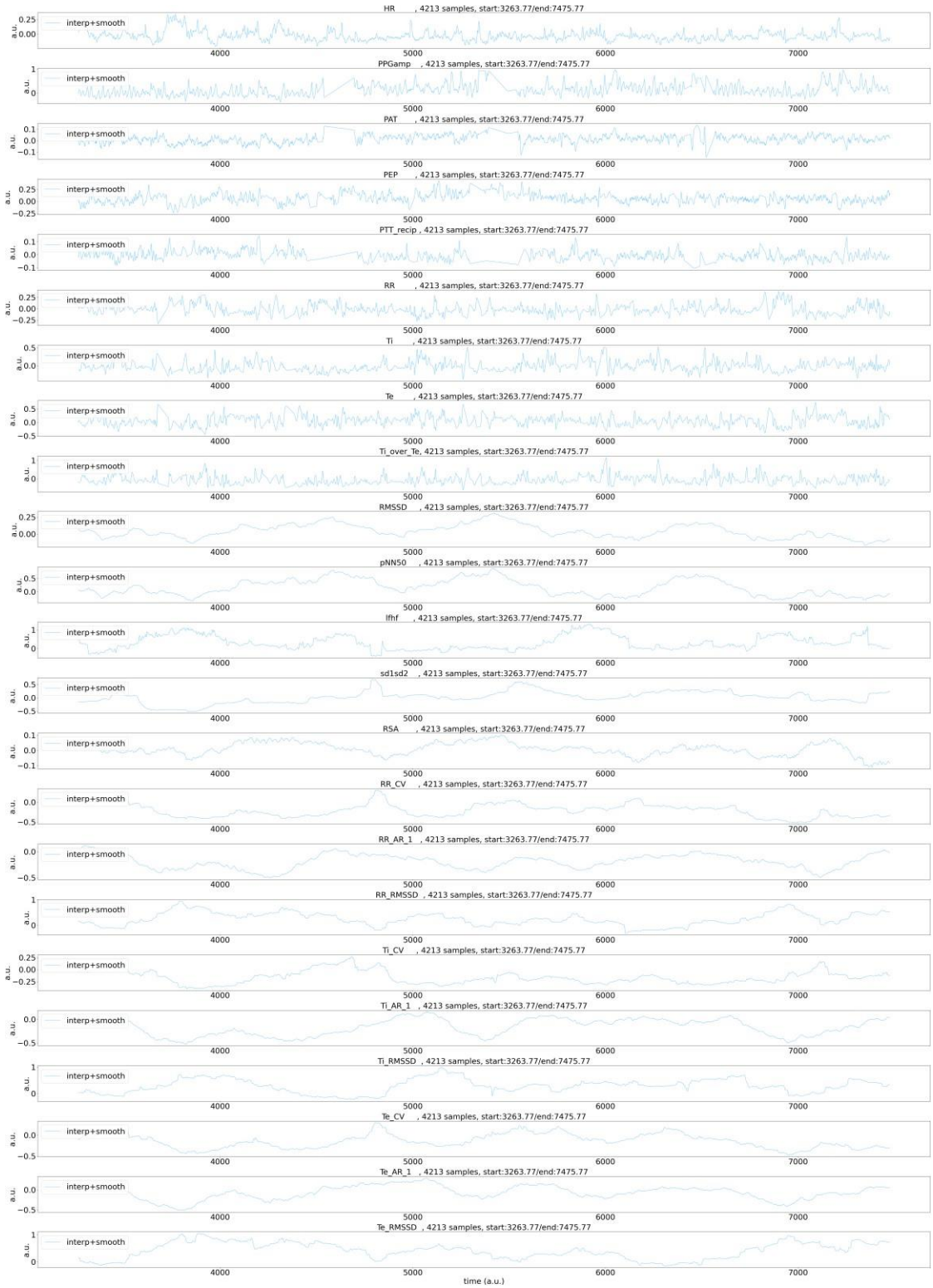

Figure 11. 23 features extracted from subject 143

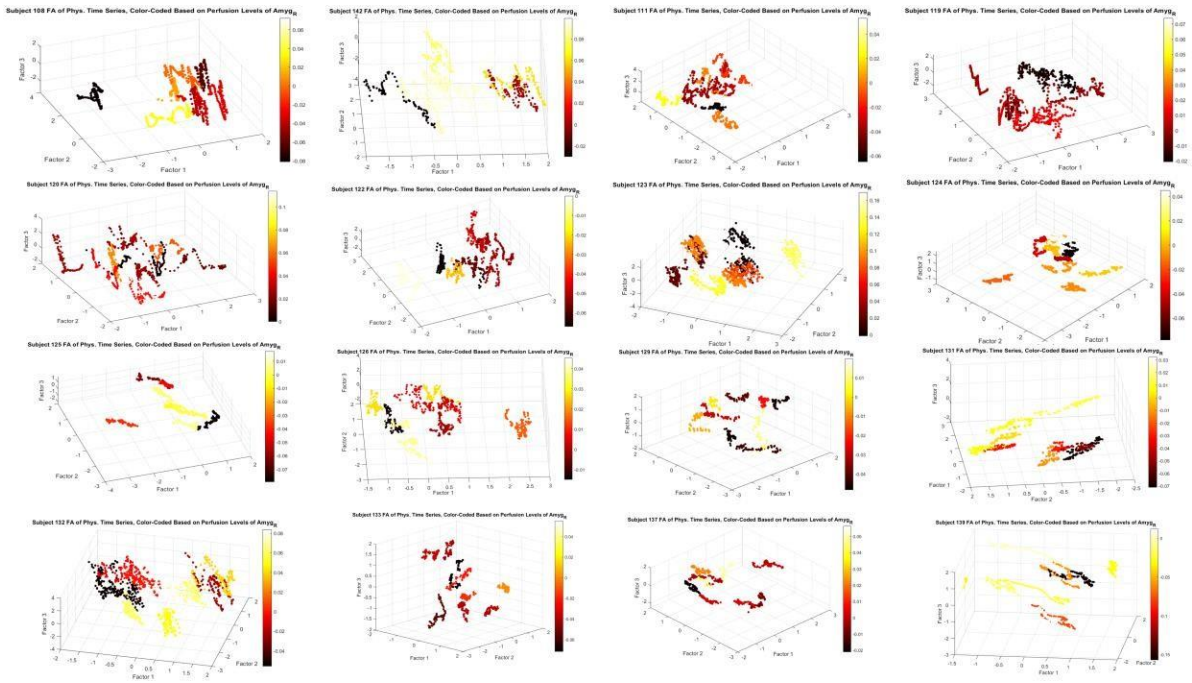

Figure 12. 23 features projected on 3 factors for all subjects

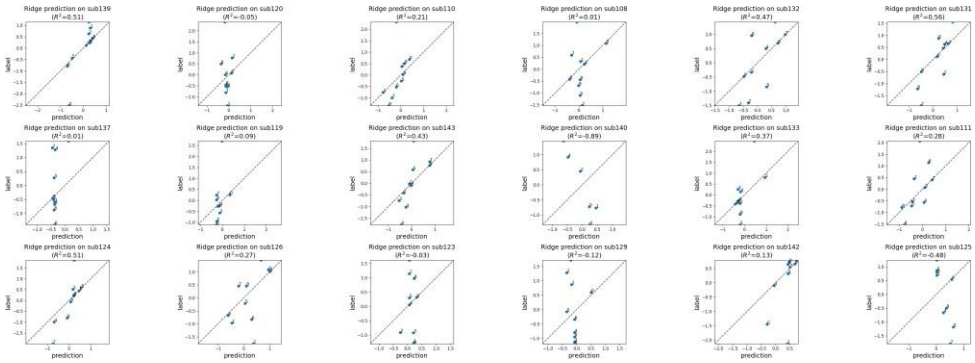

Figure 13. Ridge regression results on selected features for all subjects

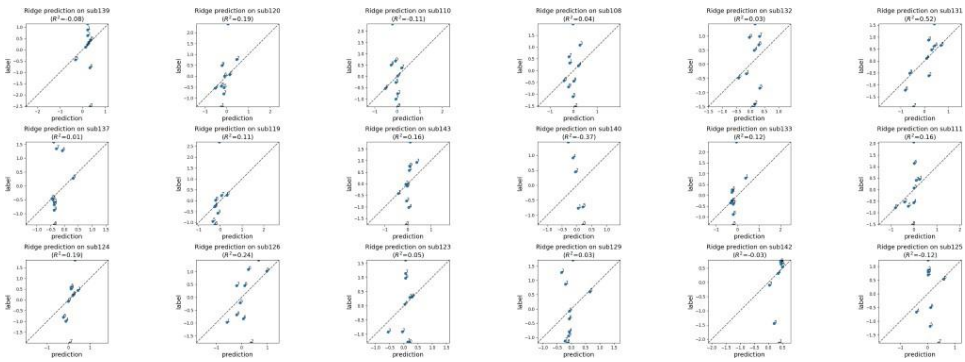

Figure 14. Ridge regression results on all features for all subjects